

EEE 554 Fall 2026

Homework 2: Handwritten Worksheet Due Saturday, September 12 on Canvas

Name: _____

1. Binomial-to-Poisson Limit (10 points)

For a binomial random variable,

$$P[X = k] = C(n, k)p^k(1-p)^{n-k}, k = 0, 1, \dots, n.$$

Consider n to infinity and p to 0 with $\lambda = np$ constant. Express the limiting PMF in terms of λ . Identify the PMF.

2. Parity Transformation of a Poisson RV (10 points)

Let X be Poisson with parameter λ , and define $Y = 1$ if X is even and $Y = -1$ if X is odd. Find the PMF of Y .

3. Best Estimate of a Bernoulli RV (10 points)

Let X be Bernoulli with parameter p , where $0 \leq p \leq 1$. Let the constant estimate be $\hat{X}$.

(a) Find the best estimate. (b) Find its mean-square error in terms of p . (c) For what p is the error maximal, and why?

4. Binomial Characteristic Function (15 points)

For a binomial random variable X : (a) calculate its characteristic function; (b) determine its expected value and variance from that characteristic function; and (c) show it becomes the characteristic function of a Poisson RV in the limit n to infinity, p to 0, with $\lambda = np$ constant.

5. Q-Function Identity (5 points)

For a PSK digital communication system, the probability of error has the form $P_e = Q[\text{square-root of } 2E/N_0]$, where N is standard normal. Prove $Q(x) = P[N > x]$.

6. Affine Transformation of a Gaussian RV (5 points)

Let X be $N(0,1)$ and $Y = 2X + \mu$. Prove that Y is $N(\mu,4)$.

7. Squared Standard Normal RV (5 points)

Let X be $N(0,1)$ and $Y = X^2$. Find $f_Y(y)$.

8. Clipped Rayleigh RV (10 points)

Let X be Rayleigh and define $Y = \min(X, x_{\max})$: $Y = X$ for $0 \leq X < x_{\max}$, and $Y = x_{\max}$ for $X \geq x_{\max}$. Find the distribution of Y .

9. Exponential Characteristic Function and Moment (10 points)

Let X be exponential with rate λ . (a) Calculate its characteristic function $\Phi_X(\omega)$. (b) Calculate the n -th moment $E[X^n]$ using the characteristic function.

Let X be $N(3,1)$. (a) Express $P[X > 8]$ as an integral, and estimate the number of realizations needed to observe 10 such events. (b) For $f_Y(y) = \exp[-(y-3)]u(y-3)$, estimate the number of realizations needed for 10 events $Y > 8$. (c) Using importance sampling, express $P[X > 8]$ as $E_Y[h(Y)]$ and find explicit h .